A Fine-tuned LLM Approach for Animal Transport Recommendations Based on Synthetic Shipment Data and EU Animal Transportation Welfare Guidelines

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Abstract

The transportation of live animals presents significant challenges in terms of logistics, animal welfare, and regulatory compliance. This paper presents a novel approach combining computer vision for animal identification with a fine-tuned large language model (tinyllama2023) to provide transport recommendations based on historical shipment data. Using ResNet50 for animal detection from photographs and TinyLlama fine-tuned on 10,000 synthetic shipment records, our system provides context-aware transport recommendations including optimal transport mode, duration, cost estimates, and welfare considerations. The model successfully integrates EU Regulation 1/2005 guidelines and scientific literature on animal transport stress factors.

1 Introduction

The global trade of live animals involves millions of shipments annually, with significant implications for animal welfare, economic efficiency, and environmental impact (5; 7). European Union legislation, particularly Council Regulation (EC) No 1/2005, establishes strict requirements for animal transport including journey times, rest periods, and vehicle specifications (6). However, implementation remains challenging due to varying interpretations and enforcement across member states.

Recent advances in machine learning offer new opportunities for optimizing animal transport logistics. Frisk et al. (7) demonstrated that route optimization can reduce transport time by up to 23% while improving animal welfare by minimizing journey duration and number of pick-up stops. Ljungberg et al. (10) showed that real-time tracking and dynamic route planning could significantly reduce queuing times at slaughterhouses.

Although cats and dogs are not classified as industrial livestock, they represent the most frequent animal transport requests from private individuals rather than commercial farming operations. Pet relocation services have grown significantly as people increasingly travel internationally with their companion animals. For simplicity in this demonstration version, cats and dogs were approximated using poultry transport parameters in the synthetic dataset,

which represents a clear limitation requiring refinement in production implementations. This demo service therefore illustrates the conceptual framework and technical capabilities rather than providing production-ready species-specific recommendations. Future versions should incorporate dedicated pet transport datasets with appropriate weight classes, airline-specific policies for in-cabin travel, and behavioral considerations distinct from livestock requirements.

This paper presents a integrated system that combines:

- Computer vision for species identification from photographs
- Historical shipment data analysis for pattern recognition
- Fine-tuned LLM for generating context-aware recommendations
- Integration of scientific welfare guidelines from EU regulations and peer-reviewed literature

2 Literature Review

2.1 Animal Welfare During Transport

Research consistently identifies journey duration and number of pick-up stops as critical welfare factors. Frisk et al. (7) found that limiting transport time to 4 hours instead of 8 reduced animal transport hours by 15.7%, though increasing operational costs by 51.5%. Mortality rates increase significantly when distances exceed 100 km (11).

Social stress from mixing unfamiliar animals represents another major welfare concern. Broom (1) documented that loading animals from multiple farms along collection routes increases agonistic behavior and disease transmission risk. The European Food Safety Authority (EFSA) emphasizes that journey conditions including road quality, driving style, and vehicle design significantly impact stress levels (13).

2.2 Transport Modes and Patterns

Analysis of EU trade data reveals distinct transport patterns by species (5). Poultry dominates intra-EU trade (97% of movements), while cattle exports to third countries represent the largest value category. Sea transport accounts for most extra-EU shipments, particularly for sheep exports to Middle Eastern and North African destinations (6).

Cockram (2) established criteria for maximum journey times based on physiological needs, recommending 8-hour limits for cattle with mandatory rest periods. However, enforcement data shows frequent violations including transport of unfit animals, overstocking, and inadequate water provision (3).

2.3 Machine Learning Applications

Route optimization models for livestock collection have shown significant potential. Gribkovskaia et al. (8) developed integer programming approaches for the livestock collection problem,

while Oppen et al. (14) applied tabu search heuristics to real-world datasets. However, these approaches typically ignore animal welfare parameters beyond basic time constraints.

Recent advances in large language models offer new capabilities for integrating diverse knowledge sources. Fine-tuned models can incorporate regulatory text, scientific literature, and historical data to generate contextually appropriate recommendations (hu2021).

3 Data Collection

3.1 Dataset Source and Augmentation

We utilized the *Transportation and Logistics Tracking Dataset* available on Kaggle (machado2024) as the foundation for our shipment records. This dataset provides a rich structure with information on delivery times, routes, and regional logistics performance across multiple sheets including on-time delivery impact, customer ratings, weather conditions, and congestion effects.

To adapt it for the specific task of animal transport recommendation, the original data was significantly augmented:

- A random weight parameter was added for each cargo, ranging from 1 kg to 1000 kg, to simulate different animal sizes and enable species classification based on typical weight ranges from veterinary literature.
- Shipment details were enriched by randomly assigning specific cities to the existing country data, creating more precise origin-destination pairs for route analysis and distance calculation.
- Transport modes were mapped to animal-appropriate categories (road, sea, air, air_accompanied) based on species type and typical shipping patterns observed in EU trade statistics.
- Welfare-related parameters including risk flags, customs delay days, and inspection types were preserved and mapped to corresponding animal transport scenarios.

This augmentation process was essential to generate the 10,000 synthetic training examples used for fine-tuning our language model, bridging the gap between general logistics data and the specialized domain of live animal transportation.

3.2 Welfare Guidelines Database

We compiled welfare guidelines from multiple sources:

- EU Regulation 1/2005 requirements for maximum journey times
- EFSA scientific opinions on transport stress factors (4)
- Peer-reviewed findings on species-specific needs (7; 13)
- Industry best practices from livestock transport carriers

Table 1: Weight ranges for animal type classification

Weight range (kg)	Animal type	Category
< 5	chicken	poultry
$5 - 20$	cat	pet
$20 - 50$	dog	pet
$50 - 100$	sheep	farm
$100 - 300$	pig	farm
$300 - 600$	cow	farm
$600 - 1000$	horse	large
> 1000	elephant	exotic

4 Methods

4.1 System Architecture

Our system comprises four integrated modules:

1. **Animal Classifier**: ResNet50 pre-trained on ImageNet, fine-tuned for domestic animal recognition
2. **Transport Advisor**: Analyzes historical shipment data to identify similar transports
3. **Welfare Guidelines**: Rule-based system incorporating scientific literature
4. **LLM Generator**: TinyLlama-1.1B fine-tuned on 10,000 prompt-completion pairs

4.2 Animal Classification

We employed ResNet50 with ImageNet weights, modified for our animal classes. Top-5 predictions are mapped to our animal taxonomy using keyword matching.

4.3 LLM Fine-tuning

We fine-tuned TinyLlama-1.1B using LoRA (Low-Rank Adaptation) (hu2021) with the following configuration:

- Rank: $r = 8$
- $\alpha = 16$
- Dropout: 0.05
- Target modules: q_proj, v_proj
- Learning rate: 2×10^{-4}
- Batch size: 2 (effective 8 with gradient accumulation)
- Epochs: 5

4.4 Prompt Engineering

Training prompts follow the structure:

```
User: How to transport a {animal} from {origin} to {destination}?

Transport options from historical data:
1. {mode}: {days} days, ${cost}, confidence {confidence}%

Animal welfare guidelines for {animal}:
- Maximum journey time: {max_hours} hours
- Requirements: {requirements}

Based on historical data and welfare guidelines...
```

5 Results

5.1 Training Performance

The fine-tuning process was completed over 5 epochs.

The relatively low training loss of 0.2917 indicates successful convergence, while the processing speed of 11.34 samples per second demonstrates efficient training. The complete 10,000-example dataset was processed at a rate of approximately 136 examples per minute over the 5-epoch training duration.

5.2 Transport Recommendations

Table 2 shows sample recommendations for different scenarios.

Table 2: Sample transport recommendations

Animal	Route	Mode	Days	Cost ($)	Confidence
Cow	Sydney→Mumbai	Sea	3.9	1,949	92%
Cow	Sydney→Mumbai	Air	0.8	9,745	45%

5.3 Model Output Example

For the query "cow from Sydney to Mumbai", the fine-tuned model generated:

> Based on historical shipment data and animal welfare guidelines, recommended best transport option: Sea transport. Expected duration and cost breakdown: 3.9 days at cost $1949. Potential risks and mitigation strategies: avoid mixing unfamiliar animals. Additional notes: Stress increases when mixed with unfamiliar animals. Mortality increases beyond 100km. Recommendations: 1. Seek

transport recommendation from professional transport agent 2. Ensure proper documentation including health certificates and customs declarations 3. Schedule transport for when expected increase in risk is below 100km 4. Monitor progress closely 5. Use customs brokerage services...

5.4 Welfare Integration

The model successfully incorporated welfare guidelines from scientific literature. For cattle shipments exceeding 8 hours, it automatically flagged welfare concerns and recommended rest stops, consistent with Frisk et al. (7) findings on journey duration impacts.

6 Discussion

6.1 Key Findings

Our results demonstrate that fine-tuned language models can effectively integrate multiple knowledge sources: historical shipment patterns, regulatory requirements, and scientific welfare guidelines. The system's recommendation for Sydney→Mumbai cattle transport (sea, 3.9 days, $1949) aligns with actual trade patterns where sea transport dominates long-distance livestock exports (6).

The high confidence (92%) for this recommendation reflects consistent patterns in the training data: cattle shipments in the 400-800kg range typically use sea transport for distances exceeding 1500km. The model correctly identified that while air transport is faster (0.8 days), its low confidence (45%) reflects limited historical precedence for such shipments.

6.2 Welfare Considerations

The model's inclusion of welfare guidelines represents a significant advance over purely logistical optimization. By incorporating Frisk et al.'s (7) findings on journey duration impacts, the system can flag welfare concerns when recommended durations exceed species-specific thresholds.

The warning about "avoid mixing unfamiliar animals" directly reflects Broom's (1) research on social stress during transport.

6.3 Limitations

Several limitations should be acknowledged:

1. **Synthetic training data**: While based on real patterns, it was randomly augmented, therefore it does not does not show real transportation patterns, and the dataset lacks the noise and variability of actual shipment records

2. **Geographic coverage**: Limited to trade routes from ((machado2024)); performance degrades for uncommon origin-destination pairs.

3. **Real-time factors**: Cannot account for current weather, political conditions, or carrier availability

4. **Welfare quantification**: Guidelines are rule-based rather than learned from welfare outcome data

6.4 Future Work

Future developments should address these limitations through:

1. Expansion of the dataset and its refinement to better meet the actual transportation requirements

2. Integration with real-time APIs for weather, customs, and carrier availability

3. Expansion to include welfare outcome data (mortality, injury rates) for more nuanced recommendations

4. Reinforcement learning from user feedback to improve recommendation quality

5. Incorporation of vehicle type specifications from USDA Animal Transportation Database

6. Better service design

7 Conclusion

This paper presents a novel approach combining computer vision, historical data analysis, and fine-tuned language models for animal transport recommendations. The system successfully identifies animals from photographs, analyzes similar shipments from historical data, and generates contextually appropriate recommendations incorporating both logistical efficiency and animal welfare guidelines from scientific literature and EU regulations.

The fine-tuned TinyLlama model demonstrates that relatively small (1.1B parameter) language models can effectively integrate diverse knowledge sources when appropriately trained on domain-specific data. This approach offers a scalable framework for developing decision support systems in regulated industries where multiple knowledge sources must be synthesized.

Future work should focus on validation with real-world shipment data, integration with real-time information sources, and expansion to additional species and trade routes.

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